

to operate between 1.9-3.6 volts only. Connecting the module to 5 volts may damage it permanently; it overheats in case of reverse connection or high-voltage connection. Connect a small capacitor (say, 10µF) between Vcc and GND leads of nRF24L01+ radio. Pin details of nRF24L01+ (shown in Fig. 5) and nRF24L01+ PA and LNA modules are the same.

The nRF24L01+ module is connected to an I2C 64x128 OLED through an ATmega328P. The humidity and temperature data received by the module is displayed on the OLED. You can also see this data on the serial monitor of Arduino IDE. All the nRF24L01+ radios are capable of transmitting six channels simultaneously. Unique pipe addresses or IDs selected here for both the transmitter and receiver modules are 0xE8E8F0FE1LL.

AVR programmer

To make this project work, you need an AVR programmer. You can make

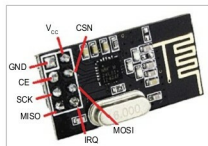


Fig. 5: Pin details of nRF24L01+ module

one yourself as described under 'Arduino as AVR Programmer' at <http://electronicsforu.com/electronics-projects/arduino-avr-programmer>

This small setup requires an Arduino board, a ZIF (zero-insertion force) socket, a blank ATmega328P and a few passive components. With these, you can create as much Arduino as you want on the fly.

First, burn the bootloader for the internal 8MHz oscillator of the ATmega328P chip. To do this, in any latest Arduino IDE, select Tools→Boards→Arduino LilyPad and then burn the Arduino bootloader onto the ATmega328P chip. LilyPad is the simplest Arduino board on internal 8MHz clock. In case the chip is already burnt as Arduino Uno (16MHz clock), it will not burn the 8MHz internal clock bootloader, unless the 16MHz quartz is connected to it. Once the internal 8MHz clock bootloader is burnt, remove the 16MHz quartz and proceed to burn the main sketch.

Extreme low power

A small lithium-ion button cell is good enough to run the small sensor hub and transmit signals up to 500 metres away for more than 25 days non-stop. On a 3.7V cell, the transmitter takes just 6-7mA current for two seconds and then sleeps for 40 seconds drawing 30µA current. The battery runs for weeks together.

The receiver draws 32mA current for ten seconds and then sleeps for 32 seconds on 8mA current.

Here's the calculation:

Current consumption (transmitter) = $(2 \times 7) + (40 \times 0.030) = 15.2\text{mA seconds}$ in 42 seconds

Therefore, in 1 second = $15.2 / 42 \text{ mA seconds} = 0.3619\text{mA seconds}$

In 1 minute = 0.3619×60

In 1 hour = $0.3619 \times 60 \times 60 \text{ mA seconds}$

In 24 hours = $0.3619 \times 60 \times 60 \times 24 \text{ mA seconds}$

= $(0.3619 \times 60 \times 60 \times 24) / 3600$

mA hours

= 8.68571mAh

On a 3.7-volt, 150mAh battery it

PARTS LIST

Semiconductors:

- IC1, IC2 - ATmega328P microcontroller
- nRF24L01+PA LNA 2.4GHz transceiver module
- nRF24L01+ 2.4GHz transceiver module
- DHT22/AM2302 temperature and relative humidity sensor module
- 4-pin 2.44cm I2C OLED module
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, ±5% carbon):

- R1-R3 - 10-kilo-ohm

Capacitors:

- C1, C3, C4 - 470µF, 16V electrolytic
- C2, C5 - 10µF, 16V electrolytic

Miscellaneous:

- CON1, CON3 - 8-pin berg strip female connector
- CON2, CON4 - 4-pin berg strip female connector
- S1, S3 - On/off toggle switch
- S2, S4 - Tactile switch
- BATT1, BATT2 - 3V-3.7V Li-ion battery
- 28-pin IC socket

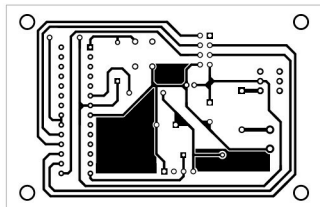


Fig. 6: Actual-size PCB layout of the transmitter

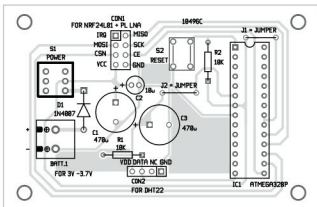


Fig. 7: Components layout for the PCB in Fig. 6